Lecture 19: AC Circuits: Theorems and Examples

Earlier we studied some network theorems, i.e.,

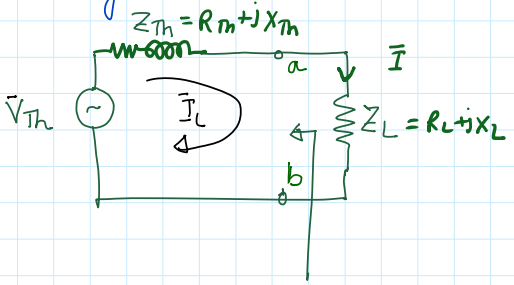
1. Thevenin's Theorem
2. Norton's Theorem
3. Superposition Theorem
4. Maximum Power Transfer Theorem
5. Delta - Star Conversion

All these are equally valid for AC circuits while using voltage and current as phasor quantities and impedances or admittances as complex quantity.

In case of maximum power transfer theorem, there is some change due to complex nature of impedance. It is given as follows.

Maximum Power Transfer Theorem:

Let Thevenin's equivalent circuit for AC circuit is given as following.



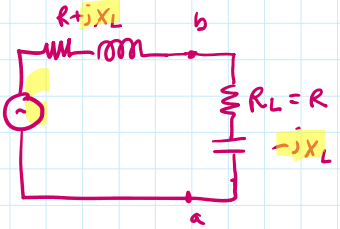
$$\bar{I}_L = \bar{I} = \left(\frac{\bar{V}_{Th}}{Z_{Th} + Z_L} \right)$$

$$P_L = I_L^2 * R_L = \left| \left(\frac{\bar{V}_{Th}}{Z_{Th} + Z_L} \right) \right|^2 * R_L$$

$$= \left| \frac{\bar{V}_{Th}}{\{(R_{Th} + R_L) + j(X_{Th} + X_L)\}} \right|^2 R_L$$

$$P_L = \frac{V_{Th}^2}{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2} * R_L$$

$$\frac{dP_L}{dR_L} = \frac{V_{Th}^2 \{ 1 * [(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2] - R_L * 2(R_{Th} + R_L) * 1}{\{(R_{Th} + R_L)^2 + (X_{Th} + X_L)^2\}^2} = 0$$



$$\Rightarrow (R_{Th} + R_L)^2 + (X_{Th} + X_L)^2 = 2R_L(R_{Th} + R_L)$$

$$(R_L + R_{Th})^2 - 2R_L(R_L + R_{Th}) + (X_L + X_{Th})^2 = 0$$

$$(R_L + R_{Th}) = \frac{2R_L \pm \sqrt{(2R_L)^2 - 4(X_L + X_{Th})^2}}{2}$$

$$R_L + R_{Th} = R_L \pm \sqrt{R_L^2 - (X_L + X_{Th})^2}$$

$$R_L + R_{Th} = R_L \pm \sqrt{R_L^2 - (X_L + X_{Th})^2}$$

$$R_{Th} = \pm\sqrt{R_L^2 - (X_L + X_{Th})^2}$$

$$R_{Th}^2 = R_L^2 - (X_L + X_{Th})^2$$

or $$R_L^2 = R_{Th}^2 + (X_L + X_{Th})^2$$

or $$R_L = \left[R_{Th}^2 + (X_L + X_{Th})^2\right]^{1/2}$$

The equation is satisfied for 2 conditions:

Case 1: When $R_L = R_{Th}$ and $X_L = -X_{Th}$

$$\Rightarrow \quad Z_L = R_L + jX_L = R_{Th} - jX_{Th} = \bar{Z}_{Th}$$

$$\boxed{Z_L = Z_{Th}^*}$$ ← condition for maximum power transfer.

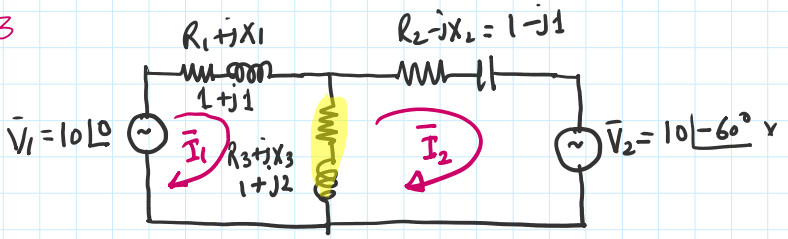
Case 2:
However if load is purely resistive, $Z_L = R_L$

Then $$\boxed{R_L = \sqrt{R_{Th}^2 + X_{Th}^2}} = |Z_{Th}|$$

In case $X_{Th} = 0$, $\boxed{R_L = R_{Th}}$.

Case 3: $Z_L = jX_L$ does not qualify for power transfer as load is purely reactive.

Example: Q 11 & 12 of Tutorial sheet 3



a) Using Mesh method, determine currents, $\bar{I}_1, \bar{I}_2, I_{Z3}$

b) Repeat (a) using Nodal method

c) Find current $\bar{I}_{Z3}$ using Thevenin's equivalent.

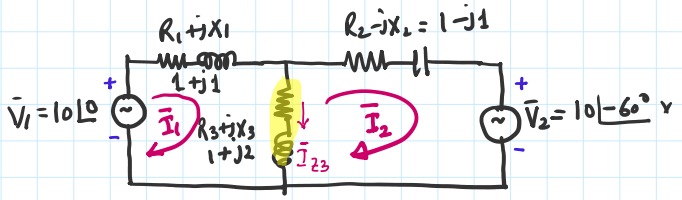
d) Calculate real, reactive, apparent powers and power factors for each source

e) What should be the value of Z_3 for maximum power transfer?

(f) Find out the maximum power and contribution of each source to this max. power.

(f) Find out the maximum power and contribution of each source to this max. power.

Solution:



(a) Mesh Method

$$-\bar{V}_1 + z_1\bar{I}_1 + z_3(\bar{I}_1 - \bar{I}_2) = 0$$

$$(z_1+z_3)\bar{I}_1 - z_3\bar{I}_2 = \bar{V}_1 \quad \text{---} \ (1)$$

$$\bar{z}_2\bar{I}_2 + \bar{V}_2 + z_3(\bar{I}_2 - \bar{I}_1) = 0$$

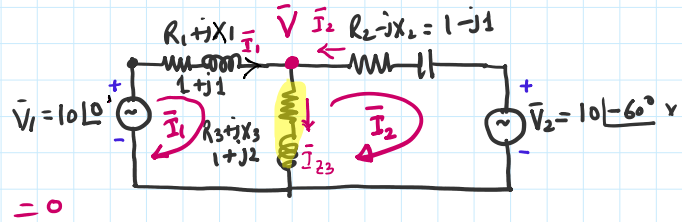
$$-z_3\bar{I}_1 + (z_2+z_3)\bar{I}_2 = -V_2 \quad \text{---} \ (2)$$

$$\begin{bmatrix} z_1+z_3 & -z_3 \\ -z_3 & z_2+z_3 \end{bmatrix}\begin{bmatrix} \bar{I}_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} \bar{V}_1 \\ -\bar{V}_2 \end{bmatrix}$$

$$\begin{bmatrix} \bar{I}_1 \\ \bar{I}_2 \end{bmatrix} = \begin{bmatrix} z_1+z_3 & -z_3 \\ -z_3 & z_2+z_3 \end{bmatrix}^{-1}\begin{bmatrix} \bar{V}_1 \\ -\bar{V}_2 \end{bmatrix} = \begin{bmatrix} 1.58\angle 60° \\ 6.05\angle 94.3° \end{bmatrix} \quad \text{Amps.}$$

$$\left.\begin{aligned} \bar{I}_1 &= 1.58\angle 60° \ \text{A} \\ \bar{I}_2 &= 6.05\angle 94.3° \ \text{A} \\ \bar{I}_{z3} &= \bar{I}_1 - \bar{I}_2 = 4.82\angle -75° \ \text{A} \end{aligned}\right\}$$

Nodal Method



KCL at mid point:

$$\frac{\bar{V}-\bar{V}_1}{\underbrace{R_1+jX_1}_{z_1}} + \frac{\bar{V}}{z_3} + \frac{\bar{V}-\bar{V}_2}{z_2} = 0$$

$$\Rightarrow \bar{V} = \frac{\left(\frac{\bar{V}_1}{z_1} + \frac{V_2}{z_2}\right)}{\left(\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right)} = \frac{\frac{10\angle 0}{1+j1} + \frac{10\angle -60°}{1-j1}}{\frac{1}{1+j1}+\frac{1}{1-j1}+\frac{1}{1+j2}} = 10.79\angle -11.56° \ \text{V}$$

$$\bar{I}_1 = \frac{\bar{V}_1 - \bar{V}}{z_1} = 1.5849\angle 60° \ \text{A}$$

$$\bar{I}_2 = \frac{\bar{V}_2 - \bar{V}}{z_2} = -6.05\angle 94.3° \ \text{A}$$

$$\bar{I}_3 = \bar{I}_1 + \bar{I}_2 = \frac{\bar{V}}{z_3} = 4.8296\angle -75° \ \text{A}$$

Source Powers

$$\bar{S}_1 = \bar{V}_1\cdot(\bar{I}_1)^* = 10\angle 0 \cdot (1.585\angle 60°)^*$$

$$= 15.85\angle -60°$$

$$= 7.92 + j(-13.73)$$

$= 7.92 + j(-13.73)$

$\bar{S}_1 = P_1 + jQ_1$

$P_1 = 7.92\ W$
$Q_1 = -13.73\ VAr$
$S_1 = 15.85 = V_1 I_1\ , VA$
$pf_1 = 0.5$ (lead)

$\bar{S}_2 = \bar{V}_2 (-\bar{I}_2)^*$, $\bar{I}_2' = -\bar{I}_2$

$= \bar{V}_2 (\bar{I}_2')^*$ $= -6.05 \angle 94.3°$

$= 10\angle -60° * 6.05 \angle 85.7°$

$-1 = 1 \angle \pm 180°$

$= 60.5 \angle 25.7°$

$6.05 \angle 94.3 - 180°$

$= 54.57 + j26.22$

$\bar{I}_2' = (6.05 \angle -85.7°)^*$

$6.05 \angle 85.7°$

$P_2 = 54.57\ W$
$Q_2 = 26\ VAr$
$S_2 = 60.5\ VA$
$pf_2 = \cos(25.7) = 0.9013$ (lag)

Powers consumed by loads:

$\bar{S}_{Z1} = \bar{V}_{Z1} \cdot (\bar{I}_{Z1})^* = (\bar{I}_{Z1} \cdot Z_1)(\bar{I}_{Z1})^*$

$= I_{Z1}^2 * Z_1 = \bar{I}_1^2 (R_1 + jX_1)$

$= (1.58)^2 (1 + j1) = 2.49 + j2.49$

$P_{Z1} + jQ_{Z1} = 2.49 + j2.49 \Rightarrow P_{Z1} = 2.49\ W;\ Q_{Z1} = 2.49\ VAr$

$\bar{S}_{Z2} = \bar{V}_{Z2} (\bar{I}_{Z2})^* = I_{Z2}^2 Z_2 = (6.05)^2 * (1 - j1) = 36.60 - j36.60$

$= P_{Z2} + jQ_{Z2} \Rightarrow P_{Z2} = 36.60\ W,\ Q_{Z2} = -36.6\ VAr$

$\bar{S}_{Z3} = \bar{V}_{Z3} \bar{I}_{Z3}^* = I_{Z3}^2 Z_3 = I_{Z3}^2 (R_3 + jX_3) = (4.82)^2 * (1 + j2)$

$P_{Z3} + jQ_{Z3} = 23.23 + j46.46 \Rightarrow P_{Z3} = 23.23\ W,\ Q_{Z3} = 46.46\ VAr$

Total power supplied by source: $\bar{S}_1 + \bar{S}_2 = 7.92 - j13.73 + 54.57 + j26.22$

$= 62.49 + j12.49$

Total power consumed by loads $= \bar{S}_{Z1} + \bar{S}_{Z2} + \bar{S}_{Z3} = 2.49 + j2.49 + 36.60 - j36.6$

$+ 23.23 + j46.46$

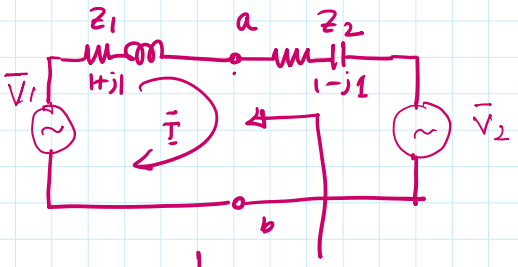
$= 62.32 + j12.35$

Thus $\bar{S}_{source} = \bar{S}_{load}$

$P_{source} = P_{load} = 62.4\ W$

$Q_{source} = Q_{load} = 12.4\ VAr$

Thevenin's Equivalent



$$\bar{V}_{Th} = \bar{V}_1 - \bar{I} Z_1 = \bar{I}_1 \bar{Z}_2 + \bar{V}_2$$

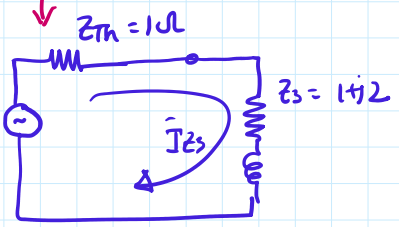
$$\bar{I} = \frac{\bar{V}_1 - \bar{V}_2}{Z_1 + Z_2} = 5\angle 60^\circ \text{ A}$$

$$\bar{V}_{Th} = 10\angle 0 - 5\angle 60^\circ * (1+j1)$$

$$\bar{V}_{Th} = 13.66\angle -30^\circ \text{ V}$$

$$Z_{Th} = Z_1 \| Z_2 = 1\,\Omega$$

$$\bar{I}_{Z3} = \frac{\bar{V}_{Th}}{Z_{Th} + Z_3} = 4.82\angle -75^\circ \text{ A}$$

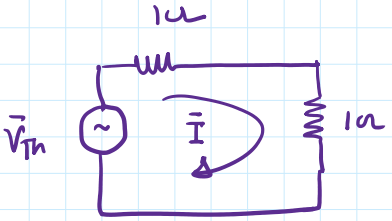


$$Z_3\big|_{\text{max power}} = Z_{Th}^* = 1\,\Omega$$

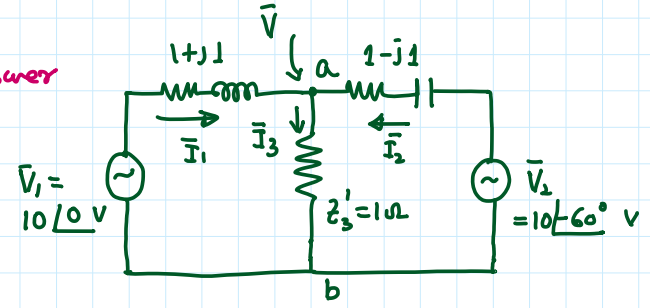
$$P_{max} = \left(\frac{\bar{V}_{Th}}{1+1}\right)^2 * 1$$

$$= \left(\frac{13.66}{2}\right)^2 * 1$$

$$\boxed{P_{max} = 46.65 \text{ W}}$$



Contribution of Each Source to Max. Power



KCL at a:

$$\frac{\bar{V} - 10\angle 0}{1+j1} + \frac{\bar{V}}{1} + \frac{\bar{V} - 10\angle -60^\circ}{1-j1} = 0$$

$$\Rightarrow \bar{V} = 6.8301\angle -30^\circ \text{ V}$$

$$\bar{I}_1 = \frac{10\angle 0 - 6.83\angle -30^\circ}{1+j1} = 3.7649\angle -5.1^\circ \text{ A}$$

$$\bar{I}_2 = \frac{10\angle -60^\circ - 6.83\angle -30^\circ}{1-j1} = 3.7649\angle -54.9^\circ \text{ A}$$

Now $\bar{S}_1 = \bar{V}_1 \bar{I}_1^* = 10\angle 0 * (3.7649\angle -5.1^\circ)^* = 37.50 + j3.35$

$\bar{S}_2 = \bar{V}_2 \bar{I}_2^* = 10\angle -60^\circ * 3.7649\angle -54.9^\circ = 37.50 - j3.35$

Power contribution from source 1, $P_1 = 37.50 - I_1^2 * r_1$

$= 37.50 - (3.7649)^2 * 1$

$= 23.32$ W

Power contribution from source 2, $P_2 = 37.50 - I_2^2 * r_2$

$= 37.50 - (3.7649)^2 * 1$

$= 23.32$ W

Total power contribution $= 23.32 + 23.32 = 46.64$ W

Total power contribution $= 23 \cdot 32 + 23 \cdot 32 = 46 \cdot 64\,W$